

Ex. No:6 Use Sleuth Kit to analyze digital evidence

Description

The Sleuth Kit (TSK) is a collection of command-line tools that allow you to analyze disk images and recover digital evidence. Here's a step-by-step guide to using Sleuth Kit on a Windows machine to analyze digital evidence:

Step 1: Install Sleuth Kit

1. Download Sleuth Kit:

- Visit the official [Sleuth Kit website](https://www.sleuthkit.org/) or https://drive.google.com/drive/u/1/folders/1ilSFY7Tqn2L7AjQGhq8yJ8kixc_xTU-v
- and download the latest version for Windows.

2. Install Sleuth Kit:

- Run the installer and follow the instructions to install Sleuth Kit on your Windows machine.

Step 2: Acquire the Disk Image

Before analysis, you need a disk image of the evidence. This can be an image of a hard drive, memory card, or any other storage device.

1. Create Disk Image:

- Use a tool like FTK Imager or dd to create a bit-by-bit copy of the storage device. Ensure the image is in a format supported by Sleuth Kit, such as .dd, .raw, .img, or .E01.
- Download the below files from the google drive
 1. 4Dell Latitude CPi.E01
 2. 4Dell Latitude CPi.E02

Step 3: Mount the Disk Image (Optional)

Mounting the disk image makes it easier to analyze the file system.

1. Mount the Image:

- You can use a tool like OSFMount to mount the image as a virtual drive on your Windows system.
- This step is optional but helps with navigating the file system easily.

Step 4: Analyze the File System

Use Sleuth Kit tools to analyze the file system and locate evidence.

1. Navigate to the Sleuth Kit Directory:

- Open the Command Prompt and navigate to the directory where Sleuth Kit is installed.

2. Identify File System Type with fsstat:

- Run the command:

arduino

```
fsstat [image file] > filesystem_info.txt
```

- This command outputs information about the file system, which is crucial for understanding the structure of the disk.

3. List Partitions with mmls:

- Run the command:

arduino

```
mmls [image file] > partitions.txt
```

- This command lists the partitions within the image file.

4. Analyze File System with fls:

- Run the command:

arduino

```
fls -r [image file] > file_list.txt
```

- This command recursively lists files and directories in the file system, showing their metadata.

5. Recover Deleted Files with icat:

- To extract a specific file, use icat:

css

```
icat [image file] [inode number] > [output file]
```

- Replace [inode number] with the inode of the file you want to recover, which you can find from the fls output.

Step 5: Analyze Metadata

Extract metadata from files to understand more about the file's history.

1. View Metadata with istat:

- Run the command:

css

```
istat [image file] [inode number] > metadata_info.txt
```

- This provides detailed information about a file, including timestamps, size, and allocation status.

Step 6: Timeline Analysis (Optional)

Creating a timeline of file activity can be crucial in an investigation.

1. Create Timeline with mactime:

- Generate a body file using fls:

css

```
fls -m / -r [image file] > body.txt
```

- Then create the timeline:

css

```
mactime -b body.txt > timeline.txt
```

- The timeline includes MAC (Modified, Accessed, Changed) times of files.

Step 7: Generate a Report

Document your findings by compiling all the data into a comprehensive report.

1. Compile the Data:

- Gather all the output files (e.g., filesystem_info.txt, partitions.txt, file_list.txt, metadata_info.txt, timeline.txt).

2. Analyze and Document:

- Review the findings, highlight important evidence, and write a report summarizing the investigation's results.

Step 8: Finalize and Store Evidence

Ensure that all evidence and reports are securely stored.

1. Archive Evidence:

- Use a secure method to archive the disk image and analysis results, ensuring integrity and confidentiality.

2. Store Securely:

- Store the archived data in a secure location, following the chain of custody procedures.

By following these steps, you can use Sleuth Kit on a Windows machine to effectively analyze digital evidence and extract crucial information for your investigation.

SCREENSHOTS :

```
C:\Users\gopi\Downloads\sleuthkit-4.14.0-win32\sleuthkit-4.14.0-win32\bin>fsstat "C:\Users\gopi\Downloads\Dell Latitude CP1.E01" > filesystem_info.txt
Cannot determine file system type

C:\Users\gopi\Downloads\sleuthkit-4.14.0-win32\sleuthkit-4.14.0-win32\bin>img_stat -i list
Supported image format types:
  raw (Single or split raw file (dd))
  ewf (Expert Witness Format (EnCase))
  vmdk (Virtual Machine Disk (VMWare, Virtual Box))
  vhd (Virtual Hard Drive (Microsoft))
  logical (Logical Directory)

C:\Users\gopi\Downloads\sleuthkit-4.14.0-win32\sleuthkit-4.14.0-win32\bin>img_stat -i ewf "C:\Users\gopi\Downloads\Dell Latitude CP1.E01"
IMAGE FILE INFORMATION
-----
Image Type:                ewf

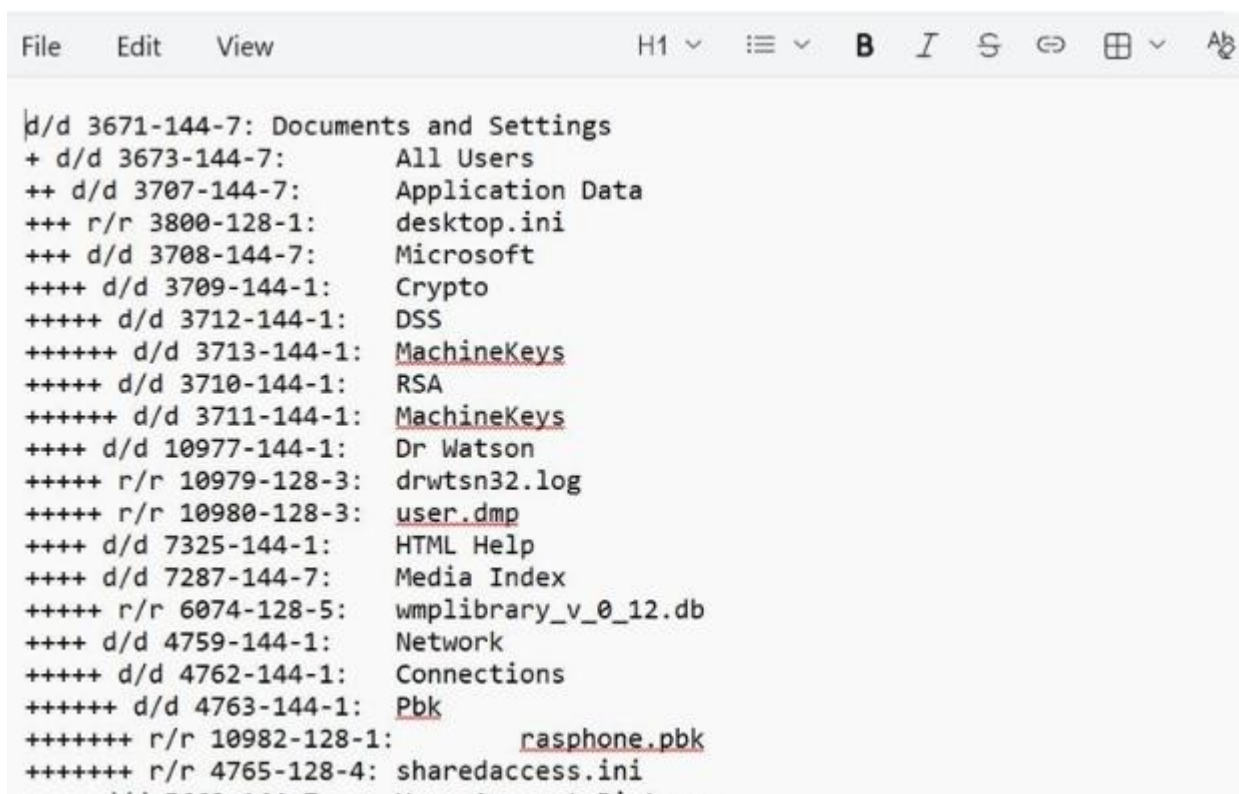
Size of data in bytes: 4871301120
Sector size:               512
MD5 hash of data:          aee4fcd9301c03b3b8454623ca261959a

C:\Users\gopi\Downloads\sleuthkit-4.14.0-win32\sleuthkit-4.14.0-win32\bin>mmls -i ewf "C:\Users\gopi\Downloads\Dell Latitude CP1.E01"
DOS Partition Table
Offset Sector: 0
Units are in 512-byte sectors

   Slot   Start      End          Length    Description
000: Meta  0000000000  0000000000  0000000001  Primary Table (#0)
001: ----- 0000000000  0000000063  0000000063  Unallocated
002: 000:000 0000000063  0009510479  0009510417  NTFS / exFAT (0x07)
003:      0009510480  0009514259  0000033780  Unallocated

C:\Users\gopi\Downloads\sleuthkit-4.14.0-win32\sleuthkit-4.14.0-win32\bin>fsstat -i ewf -o 63 "C:\Users\gopi\Downloads\Dell Latitude CP1.E01"
FILE SYSTEM INFORMATION
-----
File System Type: NTFS
Volume Serial Number: B26CB1CE6CB18D98
OEM Name: NTFS
Version: Windows XP

METADATA INFORMATION
-----
First Cluster of MFT: 2097152
First Cluster of MFT Mirror: 4765208
```



File Edit View H1 ≡ B I S ↺ ☐ ▾ Ab

d/d 3671-144-7: Documents and Settings

- + d/d 3673-144-7: All Users
- ++ d/d 3707-144-7: Application Data
- +++ r/r 3800-128-1: desktop.ini
- +++ d/d 3708-144-7: Microsoft
- ++++ d/d 3709-144-1: Crypto
- +++++ d/d 3712-144-1: DSS
- +++++ d/d 3713-144-1: MachineKeys
- +++++ d/d 3710-144-1: RSA
- +++++ d/d 3711-144-1: MachineKeys
- ++++ d/d 10977-144-1: Dr Watson
- +++++ r/r 10979-128-3: drwtsn32.log
- +++++ r/r 10980-128-3: user.dmp
- ++++ d/d 7325-144-1: HTML Help
- ++++ d/d 7287-144-7: Media Index
- +++++ r/r 6074-128-5: wmplibrary_v_0_12.db
- ++++ d/d 4759-144-1: Network
- +++++ d/d 4762-144-1: Connections
- +++++ d/d 4763-144-1: Pbk
- +++++ r/r 10982-128-1: rasphone.pbk
- +++++ r/r 4765-128-4: sharedaccess.ini

```

C:\Users\gopi\Downloads\sleuthkit-4.14.0-win32\sleuthkit-4.14.0-win32\bin>fsstat -i ewf -o 63 "C:\Users\gopi\Downloads\Dell Latitude CP1.E01"
FILE SYSTEM INFORMATION
-----
File System Type: NTFS
Volume Serial Number: B26CB1C66CB18D98
OEM Name: NTFS
Version: Windows XP

METADATA INFORMATION
-----
First Cluster of MFT: 2097152
First Cluster of MFT Mirror: 4755208
Size of MFT Entries: 1024 bytes
Size of Index Records: 4096 bytes
Range: 0 - 12307
Root Directory: 5

CONTENT INFORMATION
-----
Sector Size: 512
Cluster Size: 512
Total Cluster Range: 0 - 9510415
Total Sector Range: 0 - 9510415

$AttrDef Attribute Values:
$STANDARD_INFORMATION (16) Size: 48-72 Flags: Resident
$ATTRIBUTE_LIST (32) Size: No Limit Flags: Non-resident
$FILE_NAME (48) Size: 68-578 Flags: Resident,Index
$OBJECT_ID (64) Size: 0-256 Flags: Resident
$SECURITY_DESCRIPTOR (80) Size: No Limit Flags: Non-resident
$VOLUME_NAME (96) Size: 2-256 Flags: Resident
$VOLUME_INFORMATION (112) Size: 12-12 Flags: Resident
$DATA (128) Size: No Limit Flags:
$INDEX_ROOT (144) Size: No Limit Flags: Resident
$INDEX_ALLOCATION (160) Size: No Limit Flags: Non-resident
$BITMAP (176) Size: No Limit Flags: Non-resident
$REPARSE_POINT (192) Size: 0-16384 Flags: Non-resident
$EA_INFORMATION (208) Size: 8-8 Flags: Resident
$EA (224) Size: 0-65536 Flags:
$LOGGED_UTILITY_STREAM (256) Size: 0-65536 Flags: Non-resident
C:\Users\gopi\Downloads\sleuthkit-4.14.0-win32\sleuthkit-4.14.0-win32\bin>fls -i ewf -o 63 -r "C:\Users\gopi\Downloads\Dell Latitude CP1.E01" > file_list.txt

```

```

Total Cluster Range: 0 - 9510415
Total Sector Range: 0 - 9510415

$AttrDef Attribute Values:
$STANDARD_INFORMATION (16) Size: 48-72 Flags: Resident
$ATTRIBUTE_LIST (32) Size: No Limit Flags: Non-resident
$FILE_NAME (48) Size: 68-578 Flags: Resident,Index
$OBJECT_ID (64) Size: 0-256 Flags: Resident
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$VOLUME_NAME (96) Size: 2-256 Flags: Resident
$VOLUME_INFORMATION (112) Size: 12-12 Flags: Resident
$DATA (128) Size: No Limit Flags:
$INDEX_ROOT (144) Size: No Limit Flags: Resident
$INDEX_ALLOCATION (160) Size: No Limit Flags: Non-resident
$BITMAP (176) Size: No Limit Flags: Non-resident
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$EA (224) Size: 0-65536 Flags:
$LOGGED_UTILITY_STREAM (256) Size: 0-65536 Flags: Non-resident

C:\Users\gopi\Downloads\sleuthkit-4.14.0-win32\sleuthkit-4.14.0-win32\bin>fls -i ewf -o 63 -r "C:\Users\gopi\Downloads\Dell Latitude CP1.E01" > file_list.txt
C:\Users\gopi\Downloads\sleuthkit-4.14.0-win32\sleuthkit-4.14.0-win32\bin>notepad file_list.txt
C:\Users\gopi\Downloads\sleuthkit-4.14.0-win32\sleuthkit-4.14.0-win32\bin>findstr /i "deleted" file_list.txt
----- File(s) containing 'deleted' items -----
C:\Users\gopi\Downloads\sleuthkit-4.14.0-win32\sleuthkit-4.14.0-win32\bin>iistat -i ewf -o 63 "C:\Users\gopi\Downloads\Dell Latitude CP1.E01" 11401-120-4 > metadata_info.txt
C:\Users\gopi\Downloads\sleuthkit-4.14.0-win32\sleuthkit-4.14.0-win32\bin>notepad metadata_info.txt
C:\Users\gopi\Downloads\sleuthkit-4.14.0-win32\sleuthkit-4.14.0-win32\bin>icat -i ewf -o 63 "C:\Users\gopi\Downloads\Dell Latitude CP1.E01" 11401-120-4 > "Deleted Items.dbx"
C:\Users\gopi\Downloads\sleuthkit-4.14.0-win32\sleuthkit-4.14.0-win32\bin>dir "Deleted Items.dbx"
Volume in drive C has no label.
Volume Serial Number is 686C-28C1

Directory of C:\Users\gopi\Downloads\sleuthkit-4.14.0-win32\sleuthkit-4.14.0-win32\bin
26-08-2026 16:32 142,036 Deleted Items.dbx
1 File(s) 142,036 bytes
0 Dir(s) 22,167,606,688 bytes free
C:\Users\gopi\Downloads\sleuthkit-4.14.0-win32\sleuthkit-4.14.0-win32\bin>

```

MFT Entry Header Values:

Entry: 11441 Sequence: 6
\$LogFile Sequence Number: 32149635
Allocated File
Links: 2

\$STANDARD_INFORMATION Attribute Values:

Flags: Archive
Owner ID: 0
Security ID: 295 (S-1-5-21-2000478354-688789844-1708537768-1003)
Created: 2004-08-21 02:48:30.556104000 (India Standard Time)
File Modified: 2004-08-21 02:48:30.566118400 (India Standard Time)
MFT Modified: 2004-08-21 02:48:30.566118400 (India Standard Time)
Accessed: 2004-08-21 02:48:30.566118400 (India Standard Time)

\$FILE_NAME Attribute Values:

Flags: Archive
Name: DELETE~1.DBX
Parent MFT Entry: 11424 Sequence: 1
Allocated Size: 0 Actual Size: 0
Created: 2004-08-21 02:48:30.556104000 (India Standard Time)
File Modified: 2004-08-21 02:48:30.556104000 (India Standard Time)
MFT Modified: 2004-08-21 02:48:30.556104000 (India Standard Time)
Accessed: 2004-08-21 02:48:30.556104000 (India Standard Time)

\$FILE_NAME Attribute Values:

Flags: Archive
Name: Deleted Items.dbx
Parent MFT Entry: 11424 Sequence: 1
Allocated Size: 0 Actual Size: 0
Created: 2004-08-21 02:48:30.556104000 (India Standard Time)
File Modified: 2004-08-21 02:48:30.556104000 (India Standard Time)
MFT Modified: 2004-08-21 02:48:30.556104000 (India Standard Time)
Accessed: 2004-08-21 02:48:30.556104000 (India Standard Time)

Attributes:

Type: \$STANDARD_INFORMATION (16-0) Name: N/A Resident size: 72
Type: \$FILE_NAME (48-3) Name: N/A Resident size: 90
Type: \$FILE_NAME (48-2) Name: N/A Resident size: 100
Type: \$DATA (128-4) Name: N/A Non-Resident size: 142036 init_size: 126976
3370582 3370583 3370584 3370585 3370586 3370587 3370588 3370589
3370590 3370591 3370592 3370593 3370594 3370595 3370596 3370597
3370598 3370599 3370600 3370601 3370602 3370603 3662487 3662488
3662489 3662490 3662491 3662492 3662493 3662494 3662495 3662496
3662497 3662498 3662499 3662500 3662501 3662502 3662503 3662504
3662505 3662506 3662507 3662508 3662509 3662510 3662511 3662512
3662513 3662514 3662515 3662516 3662517 3662518 3662519 3662520
3662521 3662522 3662523 3662524 3662525 3662526 3662527 3662528
3662529 3662530 3662531 3662532 3662533 3662534 3662535 3662536
3662537 3662538 3662539 3662540 3662541 3662542 3662543 3662544
3662545 3662546 3662547 3662548 3662549 3662550 3662551 3662552
3662553 3662554 3662555 3662556 3662557 3662558 3662559 3662560
3662561 3662562 3662563 3662564 3662565 3662566 3662567 3662568
3662569 3662570 3662571 3662572 3662573 3662574 3662575 3662576
3662577 3662578 3662579 3662580 3662581 3662582 3662583 3662584
3662585 3662586 3662587 3662588 3662589 3662590 3662591 3662592
3662593 3662594 3662595 3662596 3662597 3662598 3662599 3662600
3662601 3662602 3662603 3662604 3662605 3662606 3662607 3662608
3662609 3662610 3662611 3662612 3662613 3662614 3662615 3662616
3662617 3662618 3662619 3662620 3662621 3662622 3662623 3662624
3662625 3662626 3662627 3662628 3662629 3662630 3662631 3662632
3662633 3662634 3662635 3662636 3662637 3662638 3662639 3662640
3662641 3662642 3662643 3662644 3662645 3662646 3662647 3662648
3662649 3662650 3662651 3662652 3662653 3662654 3662655 3662656

Rubrics:

Criteria & Marks Assigned	Mark Allotted	Mark Awarded
1. GitHub Activity & Submission Regularity	3	
2. Application of Forensic Tools & Practical Execution	3	
3. Documentation & Reporting	2	
4. Engagement, Problem-Solving & Team Collaboration	2	
Total	10	

RESULT:- The Sleuth Kit successfully analyzed the digital evidence and identified the file system, partitions, metadata, and deleted files.